

Mayur Pashte

Software Engineer | Python Backend | AWS | GenAI

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Professional Summary

Software Engineer with 2+ years building Python backends and cloud-native systems on AWS. Shipped enterprise-scale platforms including an endpoint-automation system serving 150,000+ users across 6,000+ bank branches and an event-driven infrastructure-provisioning pipeline that cut server delivery from days to a hands-off automated workflow. Specialized in FastAPI, PostgreSQL, REST API design, async/event-driven architecture, and GenAI integration.

Experience

Software Engineer, Applied Cloud Computing August 2024 – Present

- Designed and shipped RESTful APIs and an internal API developer portal with authentication, rate limiting, and moderation workflows, standardizing API delivery across teams.
- Built and deployed Python backend services and REST APIs (FastAPI/Django) on AWS (Lambda, API Gateway, DynamoDB), owning features from architecture through cloud deployment.
- Implemented CI/CD pipelines for backend and frontend, automating tests, builds, and zero-downtime releases.
- Integrated GenAI features using LangChain and Google Agent Development Kit for retrieval, tool-calling, and orchestration; built Python OCR pipelines (AWS Textract) for document ingestion and structured extraction.
- Provided production support and root-cause analysis for API and automation defects, and participated in code reviews to uphold reliability and code-quality standards.

Python Intern, Squad Infotech Feb 2024 – August 2024

- Built and maintained Django views, models, and forms for internal tools.
- Debugged and tested Django-based web applications alongside senior developers.
- Strengthened REST API design, template rendering, and URL routing in Django.

Technical Skills

Languages: Python

Backend & APIs: FastAPI, REST APIs, WebSockets, Async/Event-Driven Architecture, Microservices

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, S3, EC2, Bedrock, Textract, Cognito, SNS, SES), Docker, Kubernetes, CI/CD, AWX/Ansible

Databases: PostgreSQL, DynamoDB, MongoDB

AI & GenAI: LangChain, AWS Bedrock, LLM Tool-Calling, RAG

Practices & Tools: Git, Postman

Projects

Axis End User Automation (AUEA) – Axis Bank

- Developed FastAPI-based backend services for an enterprise endpoint automation platform supporting 150,000+ users across 6,000+ branches, automating printer management, scanner diagnostics, software deployment, and self-healing operations.
- Contributed to approximately 95% automated resolution of printer-related issues by supporting printer onboarding, driver management, and device configuration workflows through PrinterIQ.
- Reduced issue resolution time from hours to minutes via automated diagnostics and self-healing remediation,

significantly reducing manual intervention in endpoint support.

- Designed and enhanced REST APIs for printer lifecycle management, device diagnostics, automation workflows, and remote remediation operations.
- Integrated Orchestrator service with PrinterIQ and a .NET Windows Agent for real-time endpoint communication using API-driven and SignalR-based workflows.
- Managed PostgreSQL databases for device, printer, and automation data; wrote and optimized SQL queries to improve backend processing performance.
- **Tech Stack:** Python, FastAPI, PostgreSQL, REST APIs, SQL, SignalR, PrinterIQ, .NET Agent, SCCM, Docker, Kubernetes, Git, Postman, Linux, Windows Services

LAAS – Infrastructure-as-Code Automation Platform – Axis Bank

- Built and owned a FastAPI callback engine driving an event-driven, 6-stage server-provisioning pipeline (AWX/Ansible + PostgreSQL), reducing provisioning from a multi-day manual runbook across 8 enterprise systems to a hands-off automated workflow with a full audit trail.
- Eliminated duplicate VM provisioning by implementing idempotency checks and PostgreSQL advisory-lock concurrency control for concurrent and retried webhook callbacks.
- Hardened reliability with a 3x exponential-backoff retry layer and 10-minute OAuth token caching, plus a background watchdog that reconciles stalled stages against AWX every 5 minutes, removing lost-callback stalls as an on-call source.
- Designed REST APIs for request intake and AWX webhook callbacks with typed validation, rejecting invalid or partial requests before persistence.
- Optimized PostgreSQL access by indexing lane/status columns, tightening connection lifecycle, and trimming large JSONB reads to keep the ops dashboard responsive under load.
- Automated post-build Qualys vulnerability scans with CSV email reporting via background threads, without impacting the core provisioning pipeline.
- **Tech Stack:** Python, FastAPI, PostgreSQL, AWX, Ansible, REST APIs, SQL, JSONB, Advisory Locks, Aurora, Docker, Linux, Windows, Git

ACC Voice Ops – AI Voice & Omnichannel Contact-Center Platform

- Developed backend and AI-integration features for a real-time voice contact-center platform bridging Twilio telephony to LLM voice agents (Google Gemini), enabling automated banking conversations in English and Hindi that deflect routine calls and escalate only qualified leads to human agents.
- Implemented LLM tool-calling handlers (30 banking and insurance functions) and an OTP verification flow (SMS via AWS SNS/Twilio and DTMF keypad capture) gating account-sensitive operations.
- Developed real-time voice over WebSockets, including barge-in interruption handling with conversation-history truncation and mu-law/PCM audio transcoding for the Gemini Live speech-to-speech path.
- Built backend REST APIs and Server-Sent Events (SSE) streams powering the operations dashboard, with resilient error handling for graceful degradation during outages.
- Built an omnichannel text pipeline (WhatsApp, email, web chat) reusing the voice tool layer with a unified conversation store and live SSE updates.
- Persisted call audit records to DynamoDB and recordings to S3 with post-call sentiment/summary enrichment; improved reliability via retry/backoff on AI calls and graceful shutdown draining active calls on deploy.
- **Tech Stack:** Node.js, TypeScript, Fastify, WebSockets, Google Gemini, Twilio, REST APIs, SSE, AWS (DynamoDB, S3, SNS, SES)

Education

University of Mumbai: Bachelor of Science in Computer Science

June 2019 – June 2023

- CGPA: 8.1/10.0
- **Coursework:** Python Full Stack Development